Operating Systems - CS(D)

Quiz No. 6

Question: Show the sequence of process execution as well as detect deadlock.

There are 7 processes: P0 to P6

There are 4 resource types:
A (12 instances), B (10 instances), C (12 instances), D (7 instances)

Process	Allocation (A B C D)	Max (A B C D)
P_0	2 1 1 0	4 2 2 2
P_1	3 2 2 1	5 3 3 3
P_2	2 1 1 2	3 2 2 3
P_3	0 1 2 0	2 2 3 1
P_4	1 2 1 1	3 3 2 2
P_5	1 1 0 1	4 2 1 2
P_6	0 1 2 1	3 3 3 2

Marks Distribution :

- Need Vector for Each Process (3.5 marks): 0.5 marks per process (7 processes).
- Available Vector (1.5 marks): For computing the initial available resources.
- Running Sequence (4 marks): For finding a safe sequence (any valid sequence is acceptable).
- Final Verdict (1 mark): For concluding the deadlock status.
- **Total Marks = 3.5 + 1.5 + 4 + 1 = 10 Marks**